Customer Churn Analysis Report Visualization

Stage 2: Data Preparation, ANN Modelling, and Customer Segmentation

Abstract

This report presents the Stage 2 deliverables for the Customer Churn Analysis project. The work focuses on data preparation, supervised modelling support, and unsupervised customer segmentation. The dataset was preprocessed through numerical scaling and categorical transformation to support downstream modelling. An Artificial Neural Network framework was prepared for churn prediction. In addition, K-means clustering was applied to identify hidden customer segments. The clustering results suggest that customer segmentation is mainly influenced by internet service type and contract structure, while tenure and senior citizen status contribute less to cluster separation. To strengthen interpretation, this report also documents the clustering workflow, model validation logic, and practical implications for customer retention strategy.

1 Introduction

Customer churn analysis is an important task in business analytics because retaining existing customers is often more cost-effective than acquiring new ones. This project aims to support churn prediction and customer segmentation through structured data preparation and machine learning analysis. The main objectives of this stage are to prepare a clean modelling dataset, support ANN-based classification, and identify meaningful customer groups using clustering methods.

Beyond prediction, segmentation provides an additional layer of value because it helps explain how different types of customers behave within the dataset. Instead of treating all customers as a single population, clustering allows the project to uncover sub-groups that may differ in service usage, contract preference, and billing patterns. These insights can be used to support more targeted business decision-making and more practical retention planning.

2 Dataset Overview

The raw dataset contains customer-related attributes and a binary churn outcome. The selected original features are shown in Table 1. The target variable is `Churn`, where `Yes` indicates customer churn and `No` indicates customer retention.

Table 1: Original features used in the project

Feature	**Type**
gender	Categorical
SeniorCitizen	Numerical / Binary indicator
Dependents	Categorical
tenure	Numerical
PhoneService	Categorical
MultipleLines	Categorical
InternetService	Categorical
Contract	Categorical
MonthlyCharges	Numerical
Churn	Target label

The raw data was first inspected to ensure that all variables were available and correctly formatted for downstream preprocessing. After inspection, the processed dataset was saved in a structured form suitable for both supervised and unsupervised learning. This was important because clustering requires a purely numerical input space, while ANN modelling also benefits from scaled and consistently encoded features.

3 Data Preparation

The dataset was preprocessed to ensure compatibility with ANN modelling and clustering analysis. The target variable `Churn` was mapped into binary form. Specifically, `Yes` was encoded as 1 and `No` was encoded as 0.

A copy of the processed dataset was also used during clustering analysis. This is a good analytical practice because it preserves the clean modelling dataset while allowing new derived variables such as cluster labels to be added without altering the original structure. As a result, comparisons between original features and clustering outcomes remain easier to manage and interpret.

3.1 Numerical Features

The numerical features included:

- `SeniorCitizen`
- `tenure`
- `MonthlyCharges`

These features were transformed using `MinMaxScaler`. This process normalised the values into a comparable range and improved suitability for ANN training and

distance-based clustering. Scaling was particularly important for K-means because the algorithm is sensitive to Euclidean distance, meaning variables with larger numerical ranges could otherwise dominate the clustering process.

3.2 Categorical Features

The categorical features included:

- `gender`
- `Dependents`
- `PhoneService`
- `MultipleLines`
- `InternetService`
- `Contract`

Among these variables, `Contract` was transformed through one-hot encoding. The remaining categorical variables were converted into numerical codes for modelling. This ensured that all input features could be interpreted by machine learning models in a structured and computationally compatible format.

3.3 Train and Test Preparation

For supervised modelling, the processed dataset was split into training and testing subsets using an 80/20 ratio. This split was applied with a fixed random state to maintain reproducibility. The processed data supports both ANN-based churn prediction and clustering-based customer segmentation.

This separation supports a fair evaluation process because the model is trained on one subset and evaluated on unseen observations from the other subset. Although the clustering task itself is unsupervised, the shared preprocessing pipeline ensures consistency across all project deliverables.

4 Artificial Neural Network Support

After preprocessing, the cleaned dataset was used to support ANN model development. The ANN structure included an input layer, two hidden dense layers, and a sigmoid output layer for binary classification. The network was trained using the Adam optimiser and binary cross-entropy loss. This setup provides a suitable baseline model for churn prediction.

The ANN workflow included:

- feature-label separation
- train-test split
- model training
- prediction on the test set
- confusion matrix and classification report generation

This modelling stage demonstrates that the prepared dataset is ready for supervised learning tasks. It also confirms that the preprocessing choices made earlier were appropriate for downstream predictive modelling.

5 Customer Segmentation with K-means

In addition to churn prediction, clustering was applied to explore whether the customer base could be segmented into distinct groups. K-means clustering was selected because it is interpretable and suitable for structured tabular data. The elbow method was used to determine a suitable number of clusters, and PCA was applied for two-dimensional visualisation.

The segmentation workflow followed a clear sequence. First, the processed dataset was loaded and checked. Next, K-means was executed repeatedly across different values of k in order to compare how within-cluster variation changed. After selecting the final number of clusters, cluster labels were attached to the customer records. Finally, dimensionality reduction and distributional analysis were used to interpret the resulting groups.

5.1 Analytical Workflow

The clustering analysis can be summarised in five main steps:

1. Load the processed dataset and verify that all features are numerical and model-ready.
2. Preserve the original processed data by creating a working copy before assigning cluster labels.
3. Use the Elbow Method to compare inertia values across different cluster numbers.
4. Train the final K-means model using the selected value of k and assign each customer to a cluster.
5. Use PCA and feature-level comparisons to interpret the structure and meaning of each customer segment.

This step-by-step approach improves transparency and helps explain not only what the clustering output is, but also how the final segments were derived.

5.2 Elbow Method

Figure 1 presents the elbow method results. The inertia decreases sharply when moving from one cluster to four clusters. After that point, the rate of improvement becomes more gradual. This pattern suggests that $k = 4$ is a reasonable choice because it balances segmentation detail and model simplicity.

From an analytical perspective, inertia represents the within-cluster sum of squared distances. A lower inertia value generally indicates tighter clusters. However, increasing k will almost always reduce inertia, so the key goal is not to minimise inertia indefinitely, but to identify a point where the marginal gain becomes limited. In this case, the change in slope around four clusters supports the selection of $k = 4$ as the most practical solution.

Table 2: Feature transformation summary

Feature	Transformation	Interpretation
Churn	Binary mapping	0 = No churn, 1 = Churn
SeniorCitizen	MinMax scaling	Binary numerical indicator where 0 = Not senior citizen, 1 = Senior citizen
tenure	MinMax scaling	Continuous variable representing customer tenure
MonthlyCharges	MinMax scaling	Continuous variable representing monthly billing amount
gender	Category encoding	Encoded from original categories Female and Male
Dependents	Category encoding	Encoded from original categories No and Yes
PhoneService	Category encoding	Encoded from original categories No and Yes
MultipleLines	Category encoding	Encoded from original categories No and Yes
InternetService	Category encoding	Encoded from original categories such as DSL and Fiber optic
Contract	One-hot encoding	Expanded into `Contract_Month-to-month` and `Contract_One year` according to the observed dataset categories

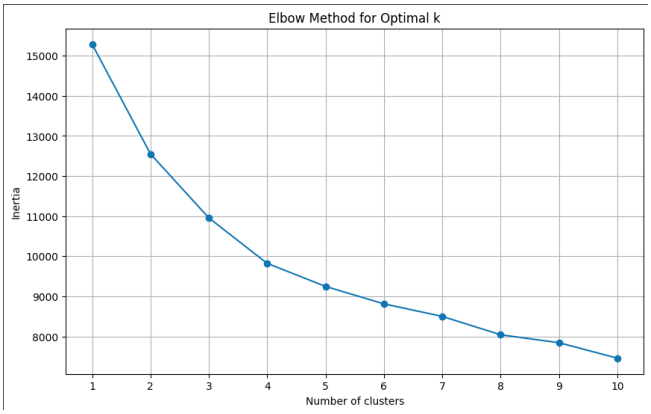

Figure 1: Elbow method for selecting the optimal number of clusters

5.3 Cluster Validation

In addition to the Elbow Method, the silhouette score was used as a supporting validation measure. The clustering result produced a silhouette score of 0.192. This value suggests that the customer groups are meaningfully separated, although some overlap remains in the feature space.

The silhouette score ranges from -1 to 1. Higher values indicate that points are more similar to their own cluster than to neighbouring clusters. In practical customer datasets, a moderate score is common because human behaviour patterns often contain natural overlap. Therefore, the score of 0.192 should not be interpreted as a failure, but rather as an indication that the clusters capture useful structure within a complex behavioural dataset.

More importantly, the silhouette result complements the Elbow Method. While the Elbow Method focuses on internal compactness through inertia, the silhouette score provides an additional perspective by evaluating separation between clusters. Together, these two criteria strengthen the decision to use four clusters.

5.4 PCA Cluster Visualisation

Figure 2 shows the clustering output projected onto two principal components. The four clusters appear visually separated with limited overlap. This indicates that the K-means model successfully identified distinct customer segments in the transformed feature space.

PCA was used because the original dataset contains multiple dimensions that cannot be directly represented in a two-dimensional plot. By compressing the information into principal components, the visualisation preserves the main structure of the data while making cluster separation easier to inspect. The visible grouping in Figure 2 provides an intuitive confirmation that the segmentation is not random.

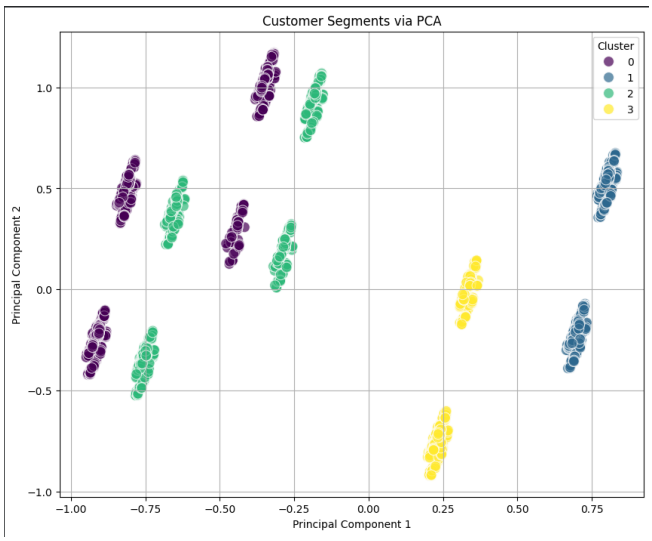

Figure 2: Customer segments visualised through PCA

5.5 Key Features Driving Segmentation

The most important variables contributing to segmentation are shown in Table 3. Internet service type was the strongest driver of cluster separation. Contract-related features also contributed substantially. In con-

trast, `tenure` and `SeniorCitizen` had no visible contribution in this output. This indicates that the clustering pattern is driven more by service choice and contractual structure than by duration of relationship or age-based demographic status.

This result is important because it shifts interpretation away from purely demographic explanations. The strongest differentiators in this customer base are operational and behavioural variables, especially the type of internet service and the structure of the customer contract. These variables appear to shape how customers group together much more strongly than variables such as tenure or senior citizen status.

Table 3: Top features driving segmentation

Feature	Importance
InternetService	0.365431
Contract_Month-to-month	0.334377
Contract_One year	0.300192
tenure	0.000000
SeniorCitizen	0.000000

5.6 Cluster Characterisation

After generating the cluster labels, each segment was examined by comparing feature behaviour across groups. This step is necessary because clustering alone only assigns labels such as Cluster 0 or Cluster 1, which have no business meaning unless they are interpreted through descriptive analysis.

Cluster interpretation was therefore based on three supporting perspectives:

- spatial separation in the PCA plot,
- feature importance results,
- comparative feature distributions across clusters.

Taken together, these perspectives suggest that the segments reflect different combinations of service type and contract preference rather than simple tenure-based groups. This makes the clustering output more useful for decision-making because the segments can be linked to actionable customer characteristics.

5.7 Cluster Distribution Analysis

Figure 3 summarises the cluster-level distributions of tenure and monthly charges. The tenure boxplots are broadly similar across the four clusters. This supports the feature importance output because tenure does not appear to be a major source of separation.

Monthly charges show moderate variation across clusters. However, the differences are less pronounced than those associated with internet service and contract structure. Therefore, monthly billing level may play a secondary role, while service type and contract pattern remain the dominant segmentation drivers.

The distribution plots are especially useful because they move the analysis from abstract clustering results to interpretable customer behaviour. Although the clusters do not differ strongly in tenure, there is some variation in monthly billing levels. This indicates that pricing characteristics may still contribute to the practical business meaning of the segments, even if they are not the strongest drivers statistically.

5.8 Strategic Interpretation

Although the clustering model was designed as an analytical task, the results also have practical strategic value. The evidence suggests that contract structure and service type are the most influential dimensions of customer grouping. This implies that retention strategy should prioritise service-related and plan-related interventions rather than relying mainly on demographic segmentation.

A reasonable practical interpretation is that customers on flexible contracts may be more sensitive to switching conditions, while customers in more structured contract groups may represent greater stability or long-term value. Similarly, service configuration appears to shape loyalty pathways more strongly than age-related or household-dependent indicators. These patterns are useful for designing customer-specific engagement strategies even when exact risk values are not directly calculated within the clustering stage.

6 Discussion

The clustering analysis provides several useful business insights. First, internet service type is the strongest factor influencing customer segmentation. This suggests that customers using different service offerings may have distinct behaviours or service needs. Second, contract structure also plays a major role. Month-to-month and one-year customers appear to form different groups, which may reflect differences in commitment level and churn risk.

By contrast, tenure does not separate the clusters in a meaningful way. The boxplots and feature importance values both support this conclusion. Senior citizen status also contributed little in this clustering output. Therefore, demographic indicators alone may be insufficient for segmenting these customers.

These findings imply that targeted retention strategies should focus more on service configuration and contract terms. Segment-specific interventions may be more effective than a single general retention policy.

The analysis also demonstrates the value of combining unsupervised learning with interpretive reporting. K-means alone provides cluster assignments, but the real business usefulness emerges when these assignments are explained through feature-level comparison and visual evidence. This project therefore shows how technical clustering outputs can be translated into practical customer intelligence.

In addition, the segmentation results support a more personalised approach to customer management. Rather than designing one retention strategy for all customers, the business can tailor communication, contract

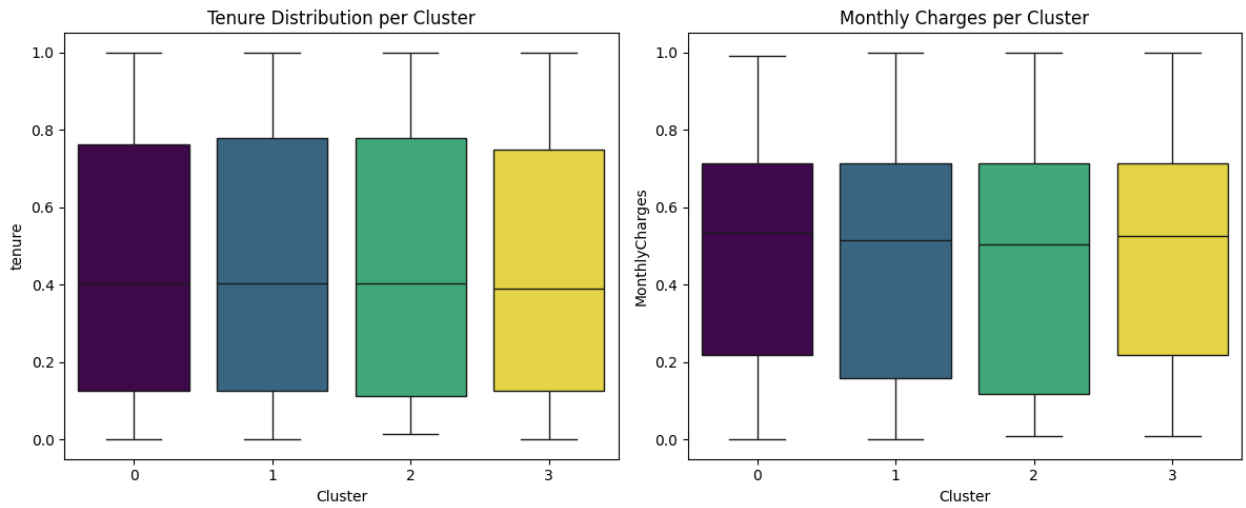

Figure 3: Cluster interpretation through feature importance and distribution analysis

Table 4: Interpretation of clustering results

Cluster	Observed pattern	Main drivers	Interpretation
0	Distinct position in PCA space with similar tenure spread to other clusters	InternetService and Contract	Represents a service and contract combination that differs clearly from the other groups
1	Clearly separated in PCA space and located on the positive side of the first principal component	InternetService and Contract	Likely corresponds to customers with a different service profile and contract structure from Cluster 0
2	Separated from Clusters 0 and 1 with comparable tenure dispersion	InternetService and Contract	Indicates another customer segment shaped mainly by service type and billing arrangement
3	Compact and distinguishable cluster with moderate monthly charge variation	InternetService and Contract	Represents a smaller but clearly defined segment within the customer base

incentives, and service offers based on segment characteristics. This improves the practical relevance of the analysis and makes the findings more actionable.

7 Conclusion

This report summarised the Stage 2 work for customer churn analysis. The dataset was prepared through structured preprocessing, including binary mapping, numerical scaling, categorical encoding, and one-hot transformation. The processed data supported ANN model development for churn prediction. In parallel, K-means clustering identified four meaningful customer segments.

The results show that customer segmentation is primarily influenced by internet service type and contract structure. Tenure and senior citizen status were less useful for distinguishing groups in this analysis. Overall, the project demonstrates that a clean and well-structured data preparation pipeline can effectively support both predictive modelling and exploratory segmentation.

More broadly, this end-to-end workflow shows how customer data can be transformed into interpretable groups that support operational decision-making. By validating cluster quality, visualising segment structure, and linking feature behaviour to business meaning, the analysis provides a stronger basis for targeted retention and customer engagement planning.

Appendix

All figures used in this report should be placed in the same folder as the `main.tex` file:

- `Elbow.png`
- `PCA.png`
- `Cluster.png`